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IMPLEMENTATION OF MACHINE LEARNING METHODS IN TRAFFIC SYSTEM AS A SOLUTION FOR MITIGATING TRAFFIC CONGESTION

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MEET OUR TEAM



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BACKGROUND

- Traffic congestion is increasing.
- Fixed-time traffic lights are inefficient.
- Manual monitoring is time-consuming.
- Intelligent transportation systems are needed.

PROBLEM STATEMENT

- Fixed Traffic Signal Durations
- Manual Vehicle Counting
- Fragmented Monitoring Data
- Prototype Performance Constraints

RESEARCH OBJECTIVES

- Detect and count vehicles using YOLO26 Nano and RF-DETR Nano.
- Determine prototype traffic signal durations using rule-based control.
- Integrate camera, GPS, detection, and traffic signal data into Firebase and the ITS Maps dashboard.
- Evaluate the functionality and stability of the ITS Maps prototype.

PROPOSED SYSTEM

Hardware:

- Raspberry Pi 5
- Raspberry Pi Camera
- GPS Module
- Prototype Traffic Signal Module
- 128 GB microSD Card
- 27 W USB-C Power Adapter

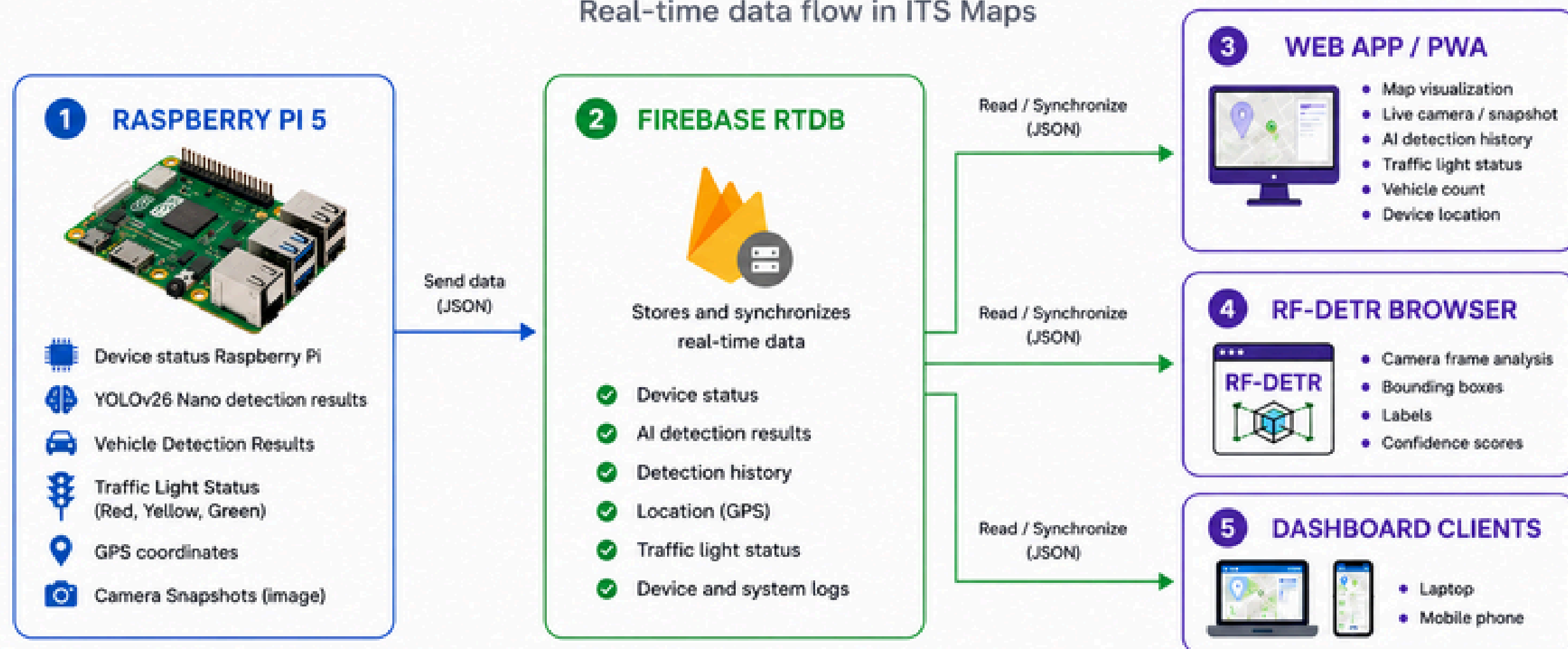
Software:

- Raspberry Pi OS 64-bit
- Scala-Based Main Application
- Python-Based Camera Service
- YOLO26 Nano on Raspberry Pi 5
- RF-DETR Nano in the Web Browser
- Firebase Realtime Database
- ITS Maps Web Dashboard

END-TO-END DATA FLOW

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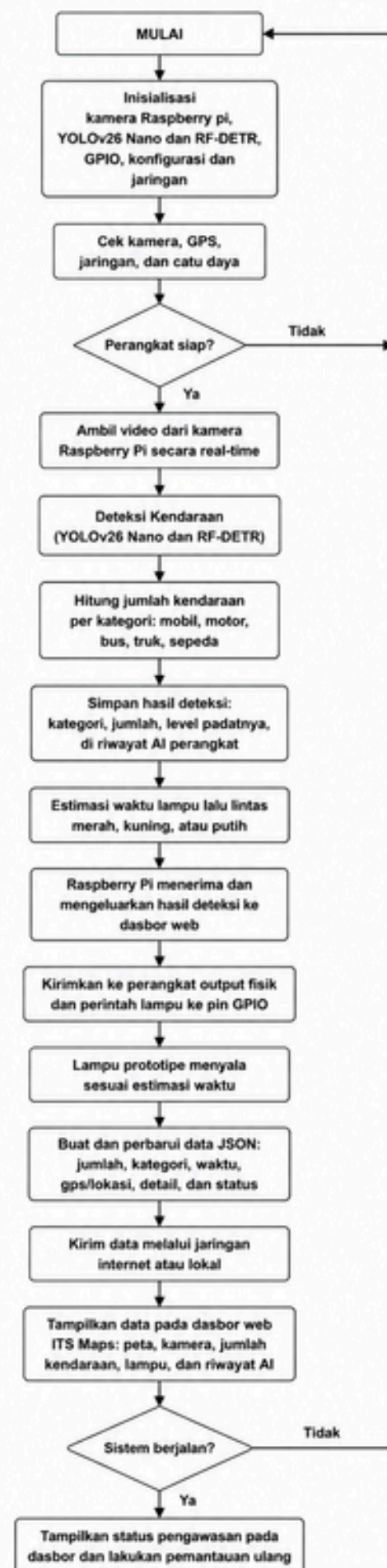
Real-time data flow in ITS Maps



DATA FORMAT

All data are structured in JSON format to ensure easy reading by dashboards, client applications, and AI modules.

FLOW CHART DESIGN



- The Raspberry Pi 5 initializes the camera, YOLO26 Nano model, GPS module, GPIO, and network connection.
- At the same time, the web dashboard loads the RF-DETR model in the browser.
- The camera captures images or videos for vehicle detection and counting.
- The detected vehicle count is used to determine traffic signal durations.
- The GPIO pins activate the prototype traffic lights.
- Detection results, location coordinates, and device status are transmitted to Firebase and displayed on the ITS Maps dashboard.
- The process is repeated continuously while the ITS Maps prototype is active.

MACHINE LEARNING METHOD

VEHICLE DETECTION METHODS

1. YOLO26 Nano

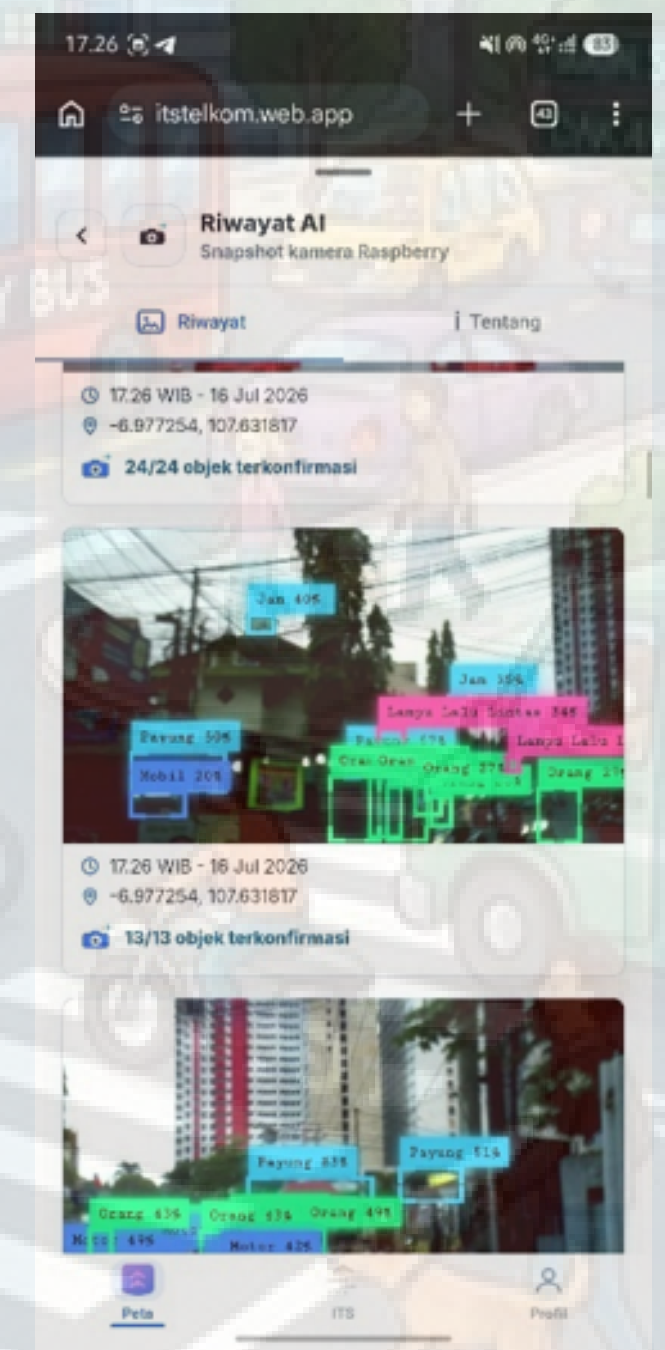
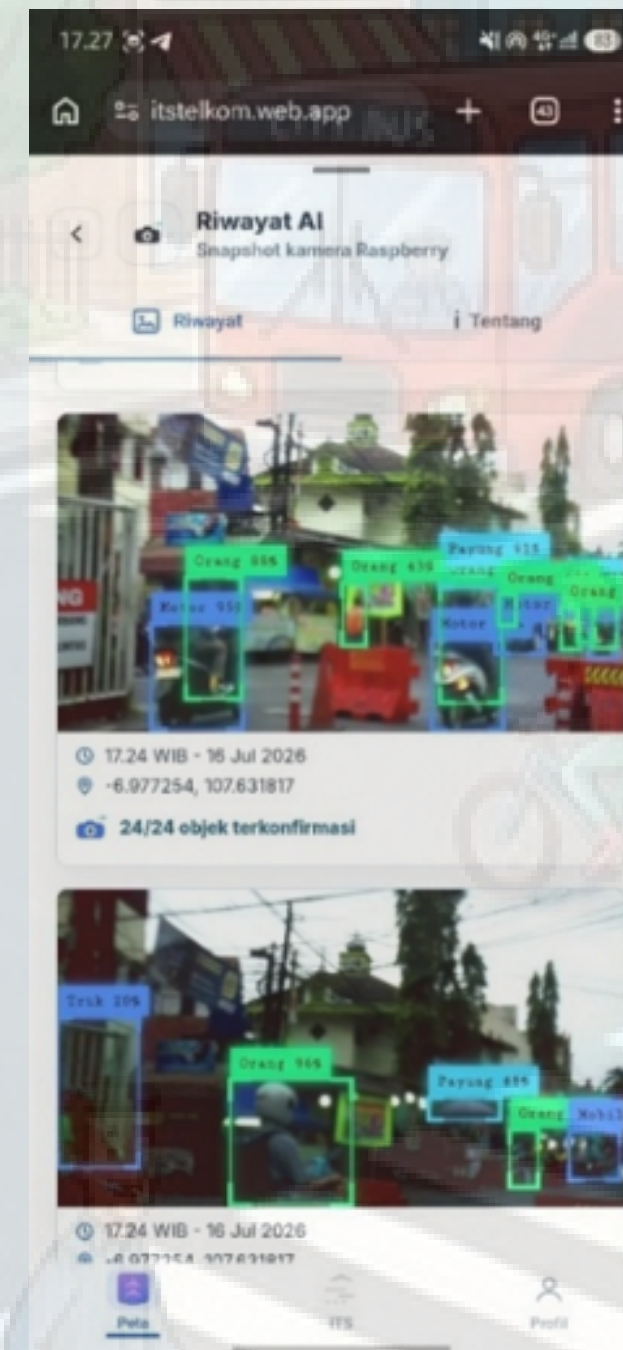
- Runs on Raspberry Pi 5
- Real-time vehicle detection
- Detects five vehicle classes
- Provides vehicle counts for traffic signal control

2. RF-DETR Nano

- Runs in the web browser
- Browser-based object detection
- Generates bounding boxes, labels detection confidence level threshold
- Displays detection results on the ITS Maps dashboard

TESTING RESULTS

1. Vehicle Detection and Counting
2. GPS Positioning
3. Prototype Traffic Signal Control
4. Firebase and Dashboard Integration
5. Prototype Functionality and Stability



CONCLUSION

ITS Maps Prototype

ITS Maps integrates vehicle detection, GPS, traffic signal control, Firebase, and web-based monitoring.

Real-Time Monitoring

Detection, GPS, traffic signal, and device data are synchronized through Firebase and displayed on the ITS Maps dashboard.

Vehicle Detection

YOLOv26 Nano and RF-DETR Nano detect and count vehicles for traffic signal control.

Recommendations

Improvements are still needed in image quality, GPS stability, network reliability, queue estimation, and multi-intersection control.

The background is a collage of various icons and illustrations related to technology, maps, and user interfaces. It includes a laptop displaying a map, a smartphone with a map app, a tablet with a line graph, a car with a blue bounding box, a cloud upload icon, a Raspberry Pi board, a blue shield with a padlock, three circular user avatars, a green shield with a checkmark, and various data visualization icons like bar charts and a camera. The entire scene is framed by blue geometric patterns in the corners.

THANK YOU

This concludes our presentation on the ITS Maps Capstone Design project. We welcome any questions, comments, and suggestions.

OFFICIAL WEB AND PROJECT RESOURCES

Scan the QR codes to access the ITS Maps application, source code repository, and technical documentation.



• ITS Maps Web

itstelkom.web.app



• Repositori GitHub

github.com/hanifasepthi/its



• Dokumentasi web

itstelkom.web.app/documentation

IMPLEMENTATION

RESEARCH LIMITATIONS

- **The system is still a single-observation-point prototype and has not been designed for long-term operation.**

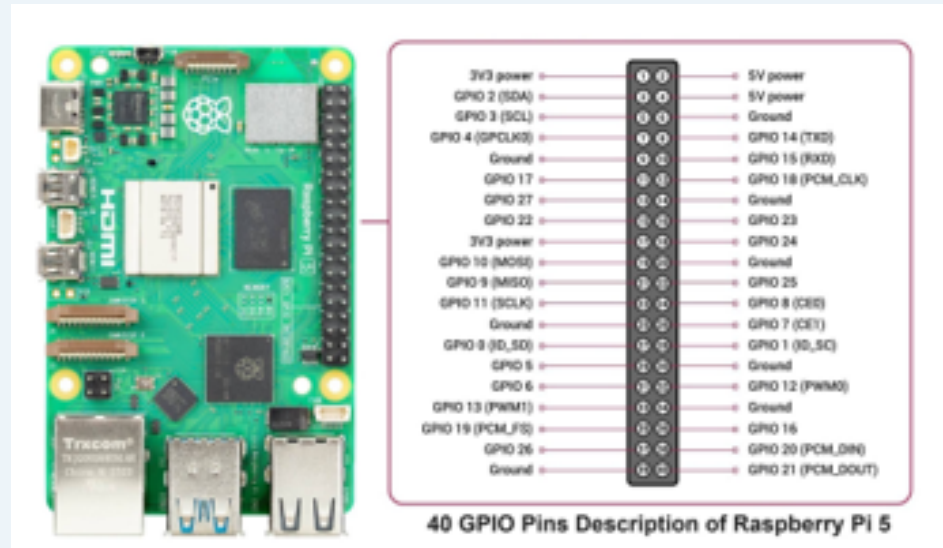
- **The YOLO26 Nano and RF-DETR models detect five vehicle classes: cars, motorcycles, buses, trucks, and bicycles.**

- Traffic signal durations are determined using predefined rules based on the number of detected vehicles. The system does not yet consider vehicle queue length or coordinate traffic signal control across multiple intersections.

- **The quality of images or videos may decrease due to lighting conditions, focus settings, object distance, camera angle, and vehicle movement.**

- The performance of GPS positioning, data transmission to Firebase, and dashboard updates is affected by power supply stability, Wi-Fi connection quality, and environmental conditions during testing.

HARDWARE



Raspberry Pi 5

- The Raspberry Pi 5 features a 2.4 GHz quad-core Arm Cortex-A76 processor, LPDDR4X memory, a 40-pin GPIO header, and a 5 V/5 A USB-C power supply.
- It serves as the main control unit for running the Scala program, Python-based camera service, GPS data acquisition, GPIO-based traffic light control, JSON data management, and communication with Firebase.



Micro SD

- A 128 GB Samsung EVO Plus microSDXC card with UHS-I, U3, V30, and A2 ratings.
- Stores the 64-bit Raspberry Pi OS, application programs, and system data.



Adaptor USB-C Raspberry Pi

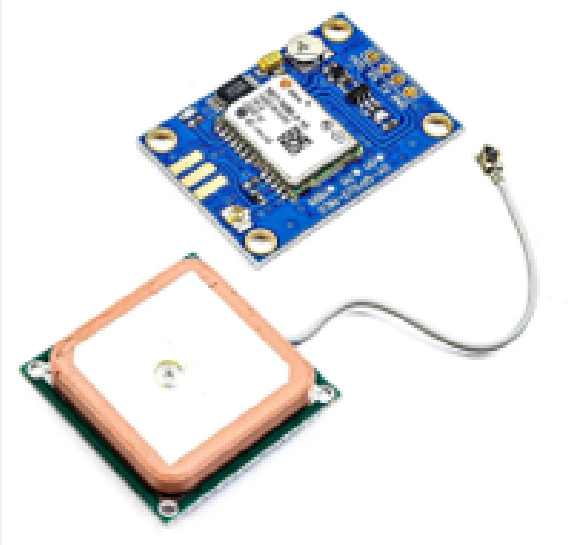
- The 27 W Raspberry Pi power adapter provides a 5.1 V/5 A USB-C output for the Raspberry Pi 5.



Camera Raspberry Pi

- The Raspberry Pi AI Camera features a 12.3 MP Sony IMX500 sensor with an integrated AI accelerator and supports 1080p30 video.
- It provides image and video input for vehicle detection using YOLO on the Raspberry Pi and RF-DETR on the web dashboard.

HARDWARE



GPS u-blox NEO M8U

- The u-blox NEO-M8U supports multiple GNSS systems, integrated inertial sensors, and position updates of up to 30 Hz.
- Provides device coordinates and testing location information.



LED traffic lights

- The red, yellow, and green LED modules are connected to Raspberry Pi GPIO pins 17, 27, and 22.
- Displays the stop, transition, and go phases of the prototype traffic signal.



Jumper wires

- Connects the Raspberry Pi 5 GPIO pins to the traffic light and GPS modules.



Raspberry Pi 5 Case and Stand

- The case protects the Raspberry Pi 5, while the mount keeps the camera stable and properly aligned.

MATHEMATICAL BASIS

Vehicle Count

Calculates the total number of detected vehicles.

$$N = n_{car} + n_{motorcycle} + n_{bus} + n_{truck} + n_{bicycle}$$

Intersection over Union (IoU)

Measures the overlap between two bounding boxes.

$$IoU(B_i, B_j) = \frac{[B_i \cap B_j]}{[B_i \cup B_j]}$$

Non-Maximum Suppression (NMS)

Reduces duplicate detections by suppressing lower-confidence overlapping boxes.

$$IoU(B_i, B_j) > \tau_{NMS} \text{ and } s_i < s_j$$

PROBLEM STATEMENT

- Fixed Traffic Signal Durations
- Manual Vehicle Counting
- Fragmented Monitoring Data
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GENERAL PROBLEM

- Traffic signal control at road intersections has not yet been able to adjust the duration of each signal phase in response to changes in vehicle volume.
- Manual vehicle counting is time-consuming and may produce inconsistent data.
- The performance of the prototype is affected by image or video quality, Wi-Fi network stability, power supply adequacy, device temperature, and GPS signal reception.

OBJECTIVES

1. Detect and count vehicles using YOLO26 Nano and RF-DETR Nano.
2. Determine prototype traffic signal durations using rule-based control.
3. Integrate camera, GNSS, detection, and traffic signal data into Firebase and the ITS Maps dashboard.
4. Evaluate the functionality and stability of the ITS Maps prototype.

SYSTEM ARCHITECTURE

Arsitektur end-to-end

